Application Developer's Guide

Provides information on using the application development interface for developing SIP and SS7 applications.

Contents

CHAPTER 1

Overview

The Application Development is a comprehensive tool to develop, deploy, and manage SIP and SS7 applications using various pre-defined application components. You can deploy your application from the Application Development interface.

Topics

This chapter includes the following topics:

About Application Development
About Application Development Applications
Using Application Development

About Application Development

The Application Development provides an environment for easily creating applications. Using the Application Development, you can also deploy these applications on the CAS.

To help you create SIP and SS7 applications, the Application Development includes various pre-defined application components organized in different modules. You can use these components to create the application logic based on requirements.

The Application Development provides a visual canvas. You can use this canvas to create your applications. It also includes an application component pallet that organizes components in various modules. To define your application logic, you drag and drop components from the relevant module on the canvas and define their relationship.

The Application Development offers the following features to the application developers:

- Easy and user-friendly interface that supports a quick and hassle-free application development environment
- Lets you create, debug, and package the application **.sar** file
- Lets you deploy the application **.sar** file on the remote CAS platform
- Lets you create database schema
- Allows you to create REST APIs for database tables supported by the SPS framework
- Supports application management tasks such as deploy, activate, deactivate, and undeploy
- Provides integrated debugging functionality that can validate the application logic and flow
- Allows multiple users to work on a single flow at the same time. Caution should be taken care in case of multiple users as the changes saved by one user may override the changes made by the other user.

About Application Development Applications

Using Application Development, you can create and deploy the SIP and SS7 (ANSI, INAP, CAP, and MAP) applications.

Using Application Development

The broad level steps for developing and deploying an application using the Application Development are listed below.

1. Install Application Development on a Linux machine. For installation instructions, see section *Getting Started*.
2. Access the Application Development interface. For details, see section *Getting Started.*
3. Create a new application. For details, see section *Developing Applications*.

4. Build the application, as required. For details, see section *Developing Applications*.
5. Deploy the application. For details, see section *Developing Applications*.

CHAPTER 2

Getting Started

This chapter introduces the Application Development interface. It also provides information on accessing the Application Development features.

Topics

This chapter includes the following topics:

Installing Application Development

Starting and Stopping Software

Accessing Application Development

Understanding the Interface

Managing Users

Managing Domains

Managing Audit Information

Tracing Calls

Understanding the Application Building Blocks

Understanding Variables

Installing Application Development

To create an application using Application Development, you must first install Application Development on a Linux machine.

Before you proceed with the installation, ensure that the following pre-requisites are met:

- Ensure that Linux RHEL version 8.4 or above is installed.
- Ensure that JDK 1.8 is installed.

The installation process requires a root user to run the commands to enable the software for a non-root user. If a non-root user runs the installer, then all the commands appear on the command prompt.

To install Application Development, follow these steps.

1 Download the **Package_x.y.z.bin** file.

2 Run the following command to install the software.

```
sh Installer_x.y.z.bin
```

The following message appears that prompts you to provide the JDK path.

Please provide JDK path:

After successful installation of the software, the installer provides the URL with the default credentials to open in web browser.

Starting and Stopping Software

After the successful installation of the software, you can start or stop it.

To start the software, follow these steps.

1 Log in to the Linux machine where is installed.

2 Enter the command.

```
service start
```

To stop the software. follow these steps.

1 Log in to the Linux machine where is installed.

2 Enter the command.

```
service stop
```

Accessing Application Development

You can access Application Development over Internet or Intranet on a stand-alone basis. Figure 1 shows the screen for Logging in to the Application Development Interface.

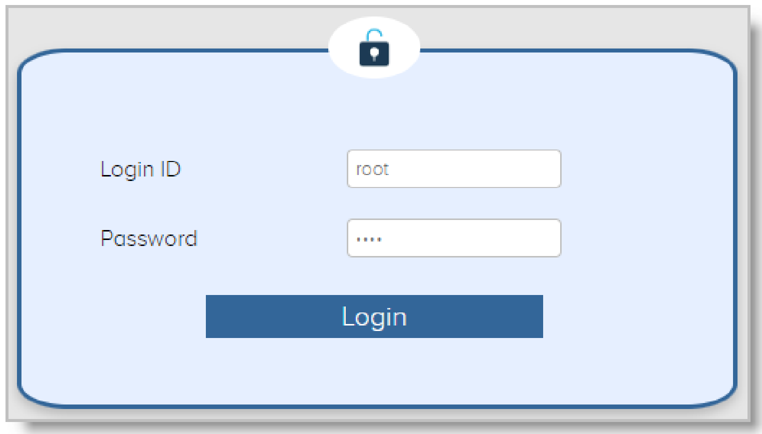


Figure 1 Logging in to the Application Development Interface

To access Application Development, follow these steps.

1. Open a web browser.
2. Enter the URL with the IP address and port number of the interface. Refer to Figure 1, Logging in to the Application Development Interface.

```
http:<serverIp:port>/UI_UIVersion/Secured/
ApplicationBuilder.html?theme=sms
```

3. Enter your login ID and password in the respective text boxes.
4. Click **Login.**

The interface displays.

To log out from the interface, click **Account** icon present on the top-right corner of the interface. Then click **Logout** button.

About Administrator Passwords

The interface prompts you to change your password in the following conditions:

When you log in to the interface for the first time.

After your old password was reset.

Figure 2 shows the screen for Changing the Password.

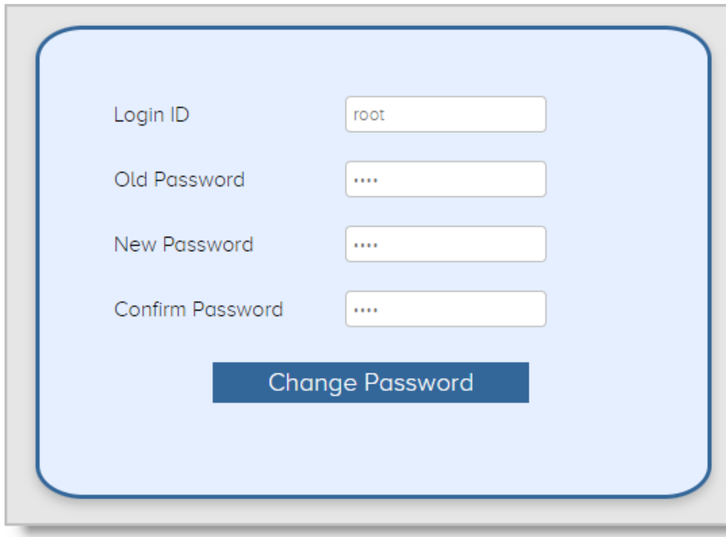


Figure 2 Changing the Password

You must enter your old and new passwords in the respective boxes. Then click Change Password. You must use the new password when you log in to the interface. The password is changed.

Once you change the password, you will be automatically logged out of the existing user session.

Reset a Root User Password

To reset a root user password, follow these steps.

1. Log in to the machine.
2. Run the following command to navigate to the database file.
3. cd $HOME/Package/WorkEnv/global/

4 Run the following command to connect to the database.

```
sqlite3 data.db
```

5 Run the following command to reset the password.

```
sqlite> UPDATE user SET password =
'6hUCqgnCtrROuuC+0B/
k8A==#noZxkRLQy6xU8o6ZHwZ6eA==', temp_password
= 1 where user_id = 'root';
```

6 Quit the command prompt.

```
sqlite> .exit
```

The root user password is reset.

Understanding the Interface

The Application Development provides a graphically-rich and user-friendly interface to Application Development functions and features. Figure 3 shows the Application Development Interface and its GUI components.

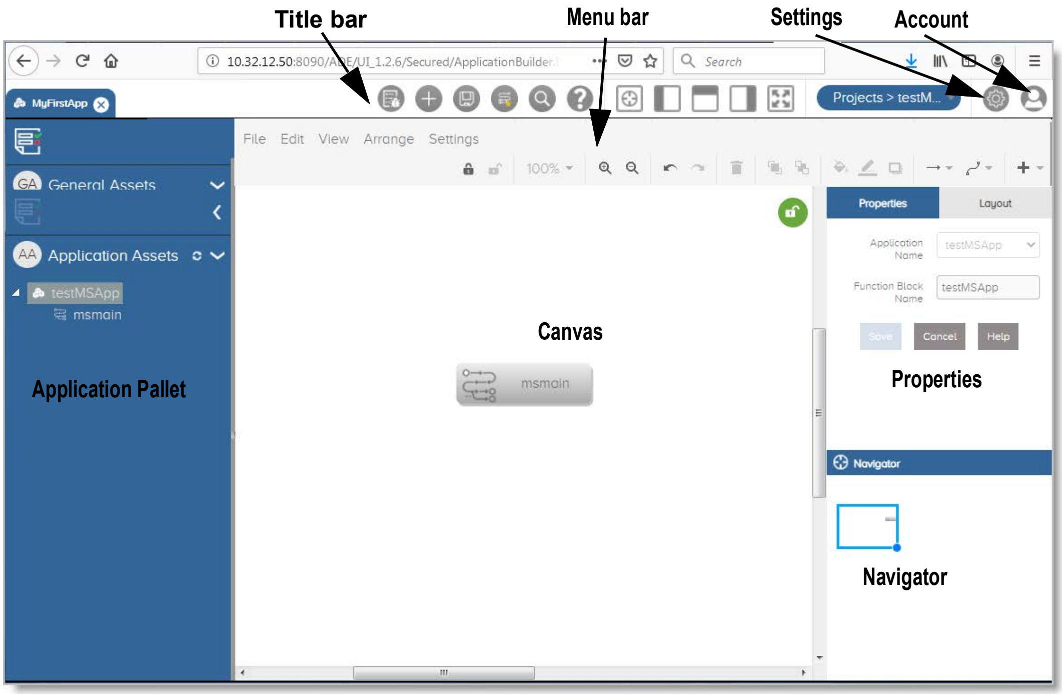


Figure 3 Application Development Interface

Table 1 describes the GUI Components of the Application Development Interface.

Table 1 GUI Components of the Application Development Interface

GUI Component	Description
Title bar	Displays the names of the open applications and other open components. It also provides icons to perform various tasks. These icons offer a shortcut to features, which you can also access using menu bar options.
Menu bar	Provides pull-down menus with options to access features and perform various tasks.
Canvas	Displays the work area where you can build and view the application logic.

Table 1 GUI Components of the Application Development Interface

GUI Component	Description
Application Pallet	Displays the application asset hierarchy and highlights the working asset. The asset might be a function block, tree, or process.
Properties	Displays and lets your configure the properties of the application and its assets. For a node, the properties panel is displayed on double-clicking the node in the canvas.
Navigator	Displays the miniature version of the application flow being created in the Canvas. You can use the Navigator to navigate within the canvas when the flow is big and it is not fitting in the visible area of the canvas.
Popup menu	Displays when you press the right mouse button in the main screen. It provides various options to perform various activities.
Dialog boxes	Displays when you select certain options from the menu bar, toolbar, popup menu, or dialog boxes. It displays information and lets you input data in fields. A dialog box has various components, including buttons, input fields, information display areas, and tabs.
Messages	Displays information, notes, and warnings on tasks being performed.

Title Bar

Figure 4 shows the Title Bar Options available in the Application Development interface.

Figure 4 Title Bar Options

Table 2 provides the Title Bar Options.

Table 2 Title Bar Icons and Description

Toolbar Icon	Description
Name	Displays the application name and other application components such as functional blocks, trees, and processes.
Debug	Debugs the business logic by providing logs generated after call execution. You need to manually copy and load the logs for debugging.
New Application	Creates a new application where you need to provide a new application name.
Save	Saves the canvas contents. It is important to save after every change to make it effective. The changes may get lost if you close the browser without saving your work.
Save As Template	Saves the current canvas content as a template.
Search	Searches the specified string in the canvas.
Navigator	Hides/unhides the navigator.
Toggle Left Panel	Hides/unhides the left pane.
Toggle Header	Hides/unhides the menu bar.
Toggle Right Panel	Hides/unhides the right pane.
Full Screen	Shows the canvas contents in the full screen mode.
Application	Lets you create a new application. It also displays a list of pre-defined applications.
Settings	Lets you create and manage domains and user accounts. It also displays audit logs and call trace files.
Account	Lets you change a password and logout of the interface.

Menu Bar

Table 3 describes the Menu Bar Options.

Table 3 Menu Bar Options

Menu Option	Description
File	Provides various options to work on a functional block, tree, template, and process.
Edit	Provides options to undo, redo, cut, copy, paste, delete, duplicate, edit, and select the components present in the canvas.
View	Provides options to show or hide the interface parts such as scroll bars, tool tips, grid, guides, connection arrows, and connection points. You can also zoom in, zoom out, scale, and adjust the size of the page view.
Arrange	Provides options to arrange the components present in the work area in a variety of ways such as bring to front, send to back, change direction, rotate, align, distribute, autosize, and clear waypoints.
Settings	Provides options to configure Application Router, Database, and package the application.
Zoom	Provides various zoom related options
Zoom In	Lets you zoom in the canvas contents
Zoom Out	Lets you zoom out the canvas contents
Undo	Undo the previous action
Redo	Redo the previous action
Delete	Delete the selected component present in the canvas
To Front	Bring the component in the front
To Back	Send the component to the back
Fill Color	Change the fill color of the selected component
Line Color	Change the line color of the selected component
Shadow	Add or remove the box shadow
Connection	Select or change the arrow style

Table 3 Menu Bar Options

Menu Option	Description
Waypoints	Select or change the arrow waypoint
Insert	Add an image or link in the canvas

Application Pallet

The application pallet displays the assets that are used to build the application. The application asset area of the application pallet shows the application hierarchy displaying the assets that have been used to create the whole application. Whenever you create an application, you will find two general assets at the top level - Function block node and Tree node. For details, see Figure 5, Application Pallet.

The Function block node provides the logical grouping of an application and may further contain one or more Function block or Tree nodes. You may start your application logic with the Tree node.

For more information on the application assets, see section Understanding the Application Building Blocks.

Figure 5 shows the Application Pallet available in the Console.

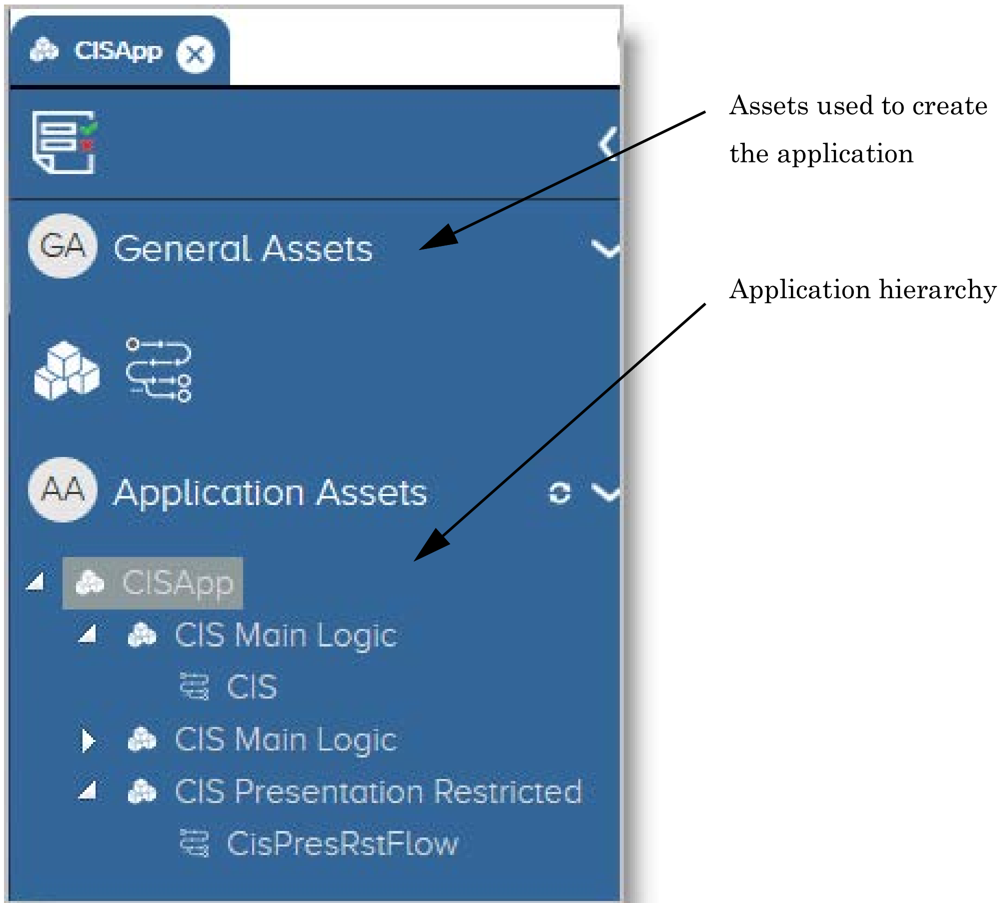


Figure 5 Application Pallet

Properties Pallet

This part of the interface lets you set the functional and visual properties of the application and its building blocks. The options shown in the properties pane keep varying depending upon the current task and selection.

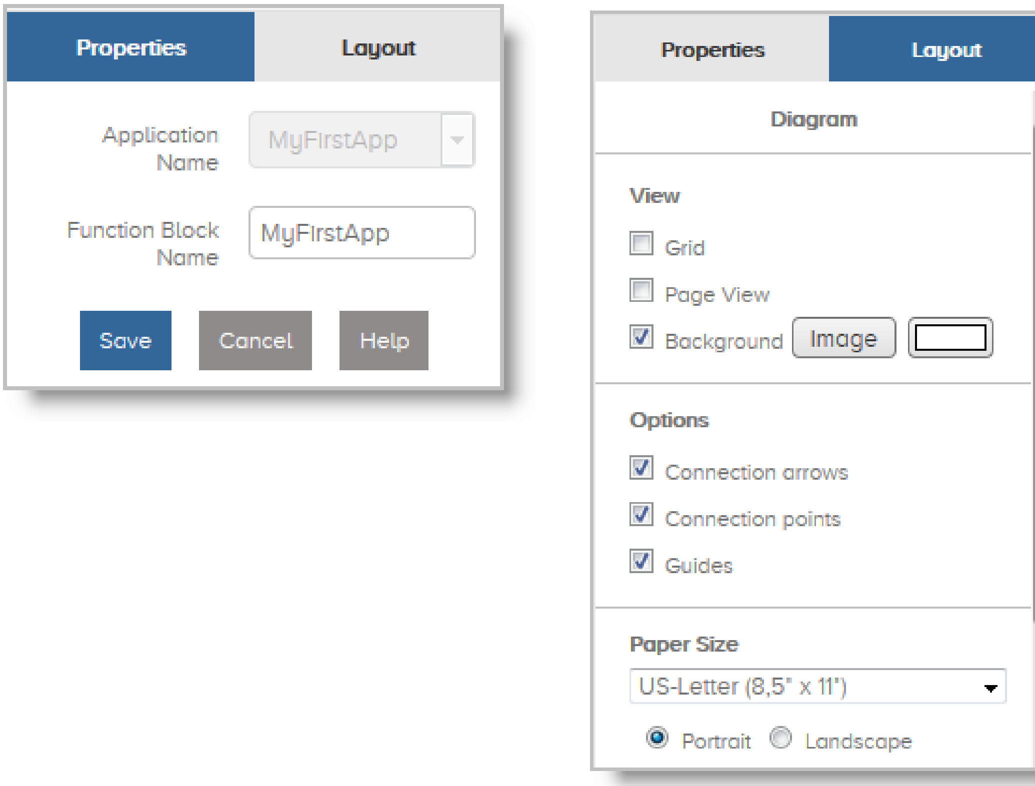


Figure 6 Properties Pallet

Canvas

This is the space where you can put together the application logic by dragging and dropping the application components. You can join them using arrows to build the complete application logic. You can scroll both horizontally and vertically in the canvas area to view the complete application flow.

Navigator

The Navigator displays the miniature version of the application flow being created in the Canvas. You can perform the following tasks:

Move to any part of the canvas to view the contents. This is especially useful when the application flow diagram becomes very big and does not fit in the visible area of the screen.

Zoom in and zoom out any part of the canvas to view the application flow diagram.

Managing Users

You can create two types of users using the interface. The admin users can manage user settings, domain settings, view audit reports, and view trace call files whereas a non-administrator user can only change the password whenever required.

This section includes the following topics:

Create a New User

Update User Details

Delete a User Record

Unlock a User Account

Force Unlock a Component

Create a New User

Figure 7 shows the pages for Creating a New User.

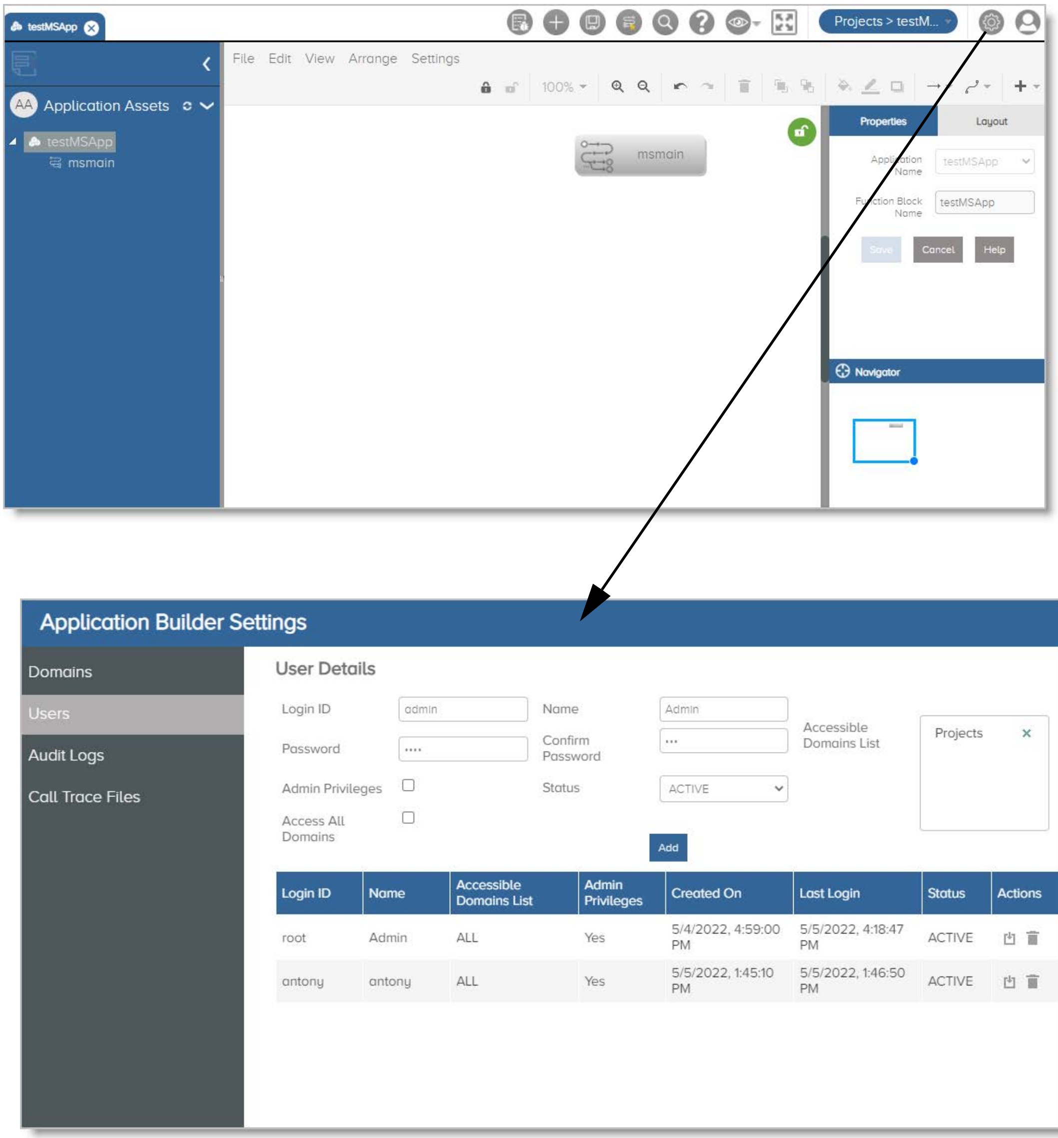


Figure 7 Creating a New User

Table 4 describes the User Details.

Table 4 User Details

Fields	Description
Login ID	Enter a unique login ID of the user.
Name	Enter a unique user name that will be used to log in to the nterface.
Password	Enter the password which is used to log in to the interface.
Confirm Password	Re-enter the password which is used to log in to the interface.
Admin Privileges	Select the check box to create an administrator. **Note:** An administrator can perform all the actions without any permission constraint.
Status	Select whether the status of the account is **Active** or **Inactive**.
Access All Domains	Select the check box to provide access for all the domains to the user. **Note:** if you select this check box, then *Accessible Domains List* field will not be visible.
Accessible Domains List	Select the domains that can be accessed by the user.

To add a new user, follow these steps.

1. Log in to the interface. Refer to Accessing Application Development.
2. Click **Settings** icon present on the top-right corner of the screen. Refer to Figure 3, Application Development Interface. The Application Builder Settings page appears.
3. By default, **Users** is selected in the left menu pane. The User Details page appears.
4. Enter the required information in the respective boxes. Refer to Table 4, User Details.
5. Click **Add**.

A new user gets added to the list.

Update User Details

You can update the user information whenever required.

Figure 8 shows the pages for Updating User Details.

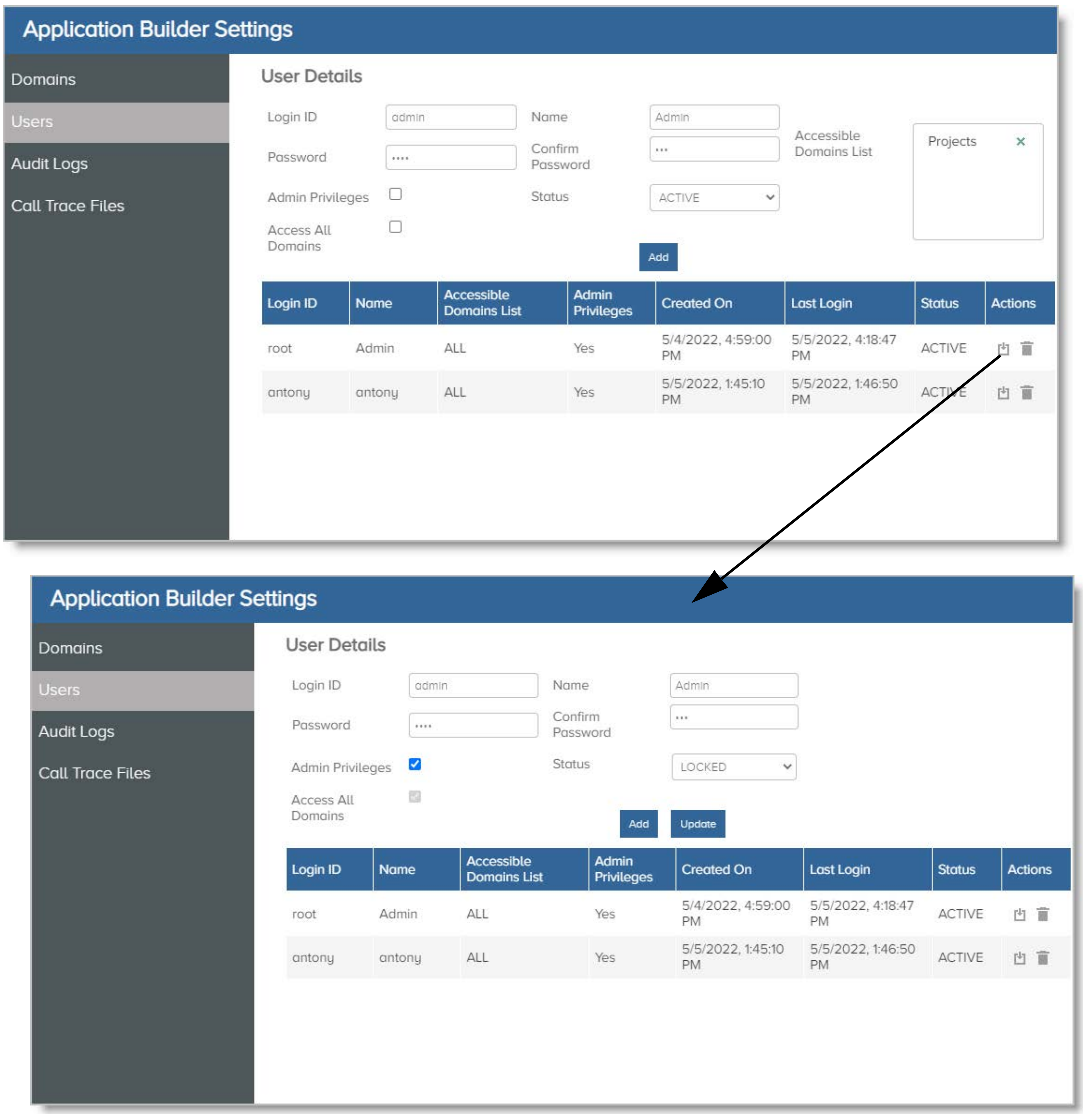


Figure 8 Updating User Details

To update the user details, follow these steps.

1. Access the User Detail page. Refer to Managing Users.
2. Identify the required user record.
3. Click **Load** icon present under the Actions column. Refer to Figure 8, Updating User Details.
4. Enter the information as required. Refer to Table 4, User Details.
5. Click **Update** button.

The identified user record gets updated.

Delete a User Record

You can delete a user account if it is no longer required.

To delete a user account, follow these steps.

1. Access the User Detail page. Refer to Managing Users.
2. Identify the required user record.
3. Click **Delete** icon present under the Actions column.

The identified user record gets deleted.

Unlock a User Account

A user with administrative privileges can unlock a user account after if it gets locked due to any reason.

To unlock a user account, follow these steps.

1. Access the User Detail page. Refer to Managing Users.
2. Identify the required user record.
3. Click **Load** icon.
4. Change the status of the user account from inactive or locked to active from the drop-down list.
5. Click **Update** button.

The identified user account gets unlocked.

Force Unlock a Component

As an administrator, you have the privilege to forcefully unlock a component and continue your work on this component.

Figure 9 shows the pages used for Unlocking a Component.

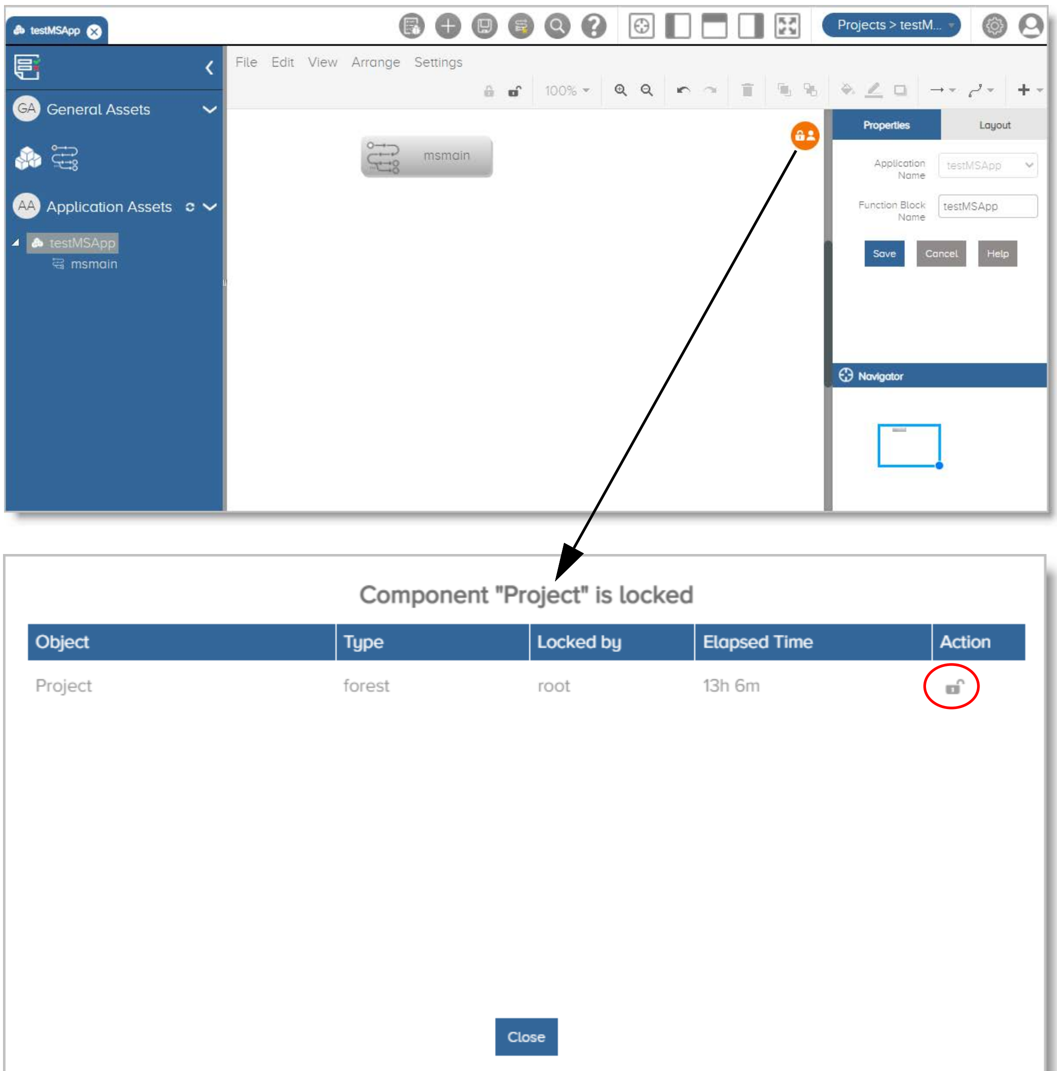


Figure 9 Unlocking a Component

To unlock a component forcefully, follow these steps.

1. Access the interface. Refer to Accessing Application Development.
2. Click **Unlock** icon. The component "name" is locked page appears. Refer to Figure 9, Unlocking a Component.
3. Click **Unlock** icon present in the Actions column.

The component gets unlocked.

Managing Domains

A domain is a user space to create single or multiple applications. When you create a new domain, you can assign it to different users who can use this space to manage their applications.

You can create a new domain and remove the domains that are no longer required.

Figure 10 shows the pages used for Creating a New Domain.

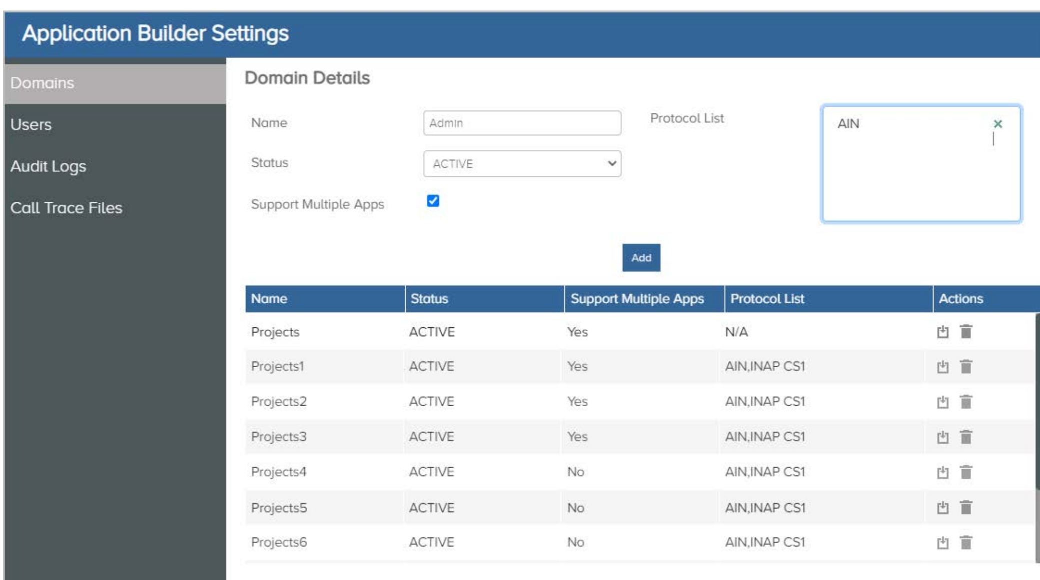


Figure 10 Creating a New Domain

Table 5 describes the Domain Details.

Table 5 Domain Details

Fields	Description
Name	Enter a unique name of the domain.
Status	Select whether the status of the domain is **Active** or **Inactive**.
Support Multiple Apps	Select the check box to enable the user to create multiple applications in this domain.
Protocol List	Select the protocols supported by the domain.

To create a new domain, follow these steps.

1. Log in to the interface. Refer to Accessing Application Development.
2. Click **Settings** icon present on the top-right corner of the screen. Refer to Figure 3, Application Development Interface. The Application Builder Settings page appears.
3. From the left menu pane, select **Domains**. The Domain Details page appears.
4. Enter the required information in the respective boxes. Refer to Table 5, Domain Details.
5. Click **Add**.

The new domain gets added to the list.

To edit a domain, identify the domain record from the list and click the **Load** icon. Edit the status of the domain and add new protocols, as required and click **Update**.

To delete a domain that is not required, identify it from the list and click **Delete** icon.

Move Domain to Another Server

Using the interface, you can move a domain present on one server to another server. For instance, if a user requires certain applications on another server, then the user can move that particular domain to another server.

To move a domain to another server, follow these steps.

1. Log in to the server.
2. Copy the domain folder from the server.
3. Paste the copied folder in the **WorkEnv** folder of the new server.

   $home_directory_of_user/Package/WorkEnv/

4. Add an entry in the **domains.csv** file present in the $home_directory_of_user/Package/WorkEnv/global directory.

   For instance, *Projects,/root/Package/WorkEnv/Projects/*

5. Restart service,

   service restart

The required domain is moved to another server.

Managing Audit Information

The audit feature is a key component of the interface. It logs all actions users perform in their sessions. These logs are available for analysis at a later time. By periodically analyzing these logs, you can identify any suspicious activities and take appropriate actions.

You can perform these audit tasks.

View Audit Report

Download Audit Report

View Audit Report

You can view audit records of the activities performed by a user for a specific duration. By default, the current date activities are displayed. You can use the scroll bar to view the records.

Figure 11 shows the pages used for Viewing Audit Report.

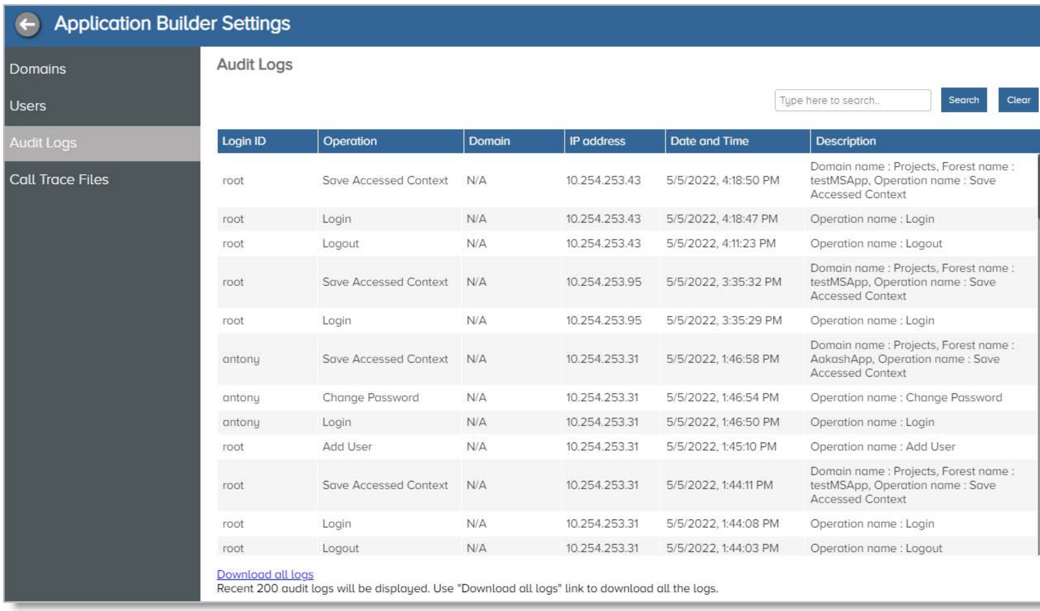


Figure 11 Viewing Audit Report

To view the audit report, follow these steps.

1. Log in to the interface. Refer to Accessing Application Development.
2. Click **Settings** icon present on the top-right corner of the screen. Refer to Figure 3, Application Development Interface. The Application Builder Settings page appears.
3. From the left pane menu, select **Audit Logs**. The Audit Logs page appears. Refer to Figure 11, Viewing Audit Report.

The audit report gets displayed on the screen.

Download Audit Report

You can export the audit records in csv format.

Figure 12 shows the pages used for Downloading an Audit Report.

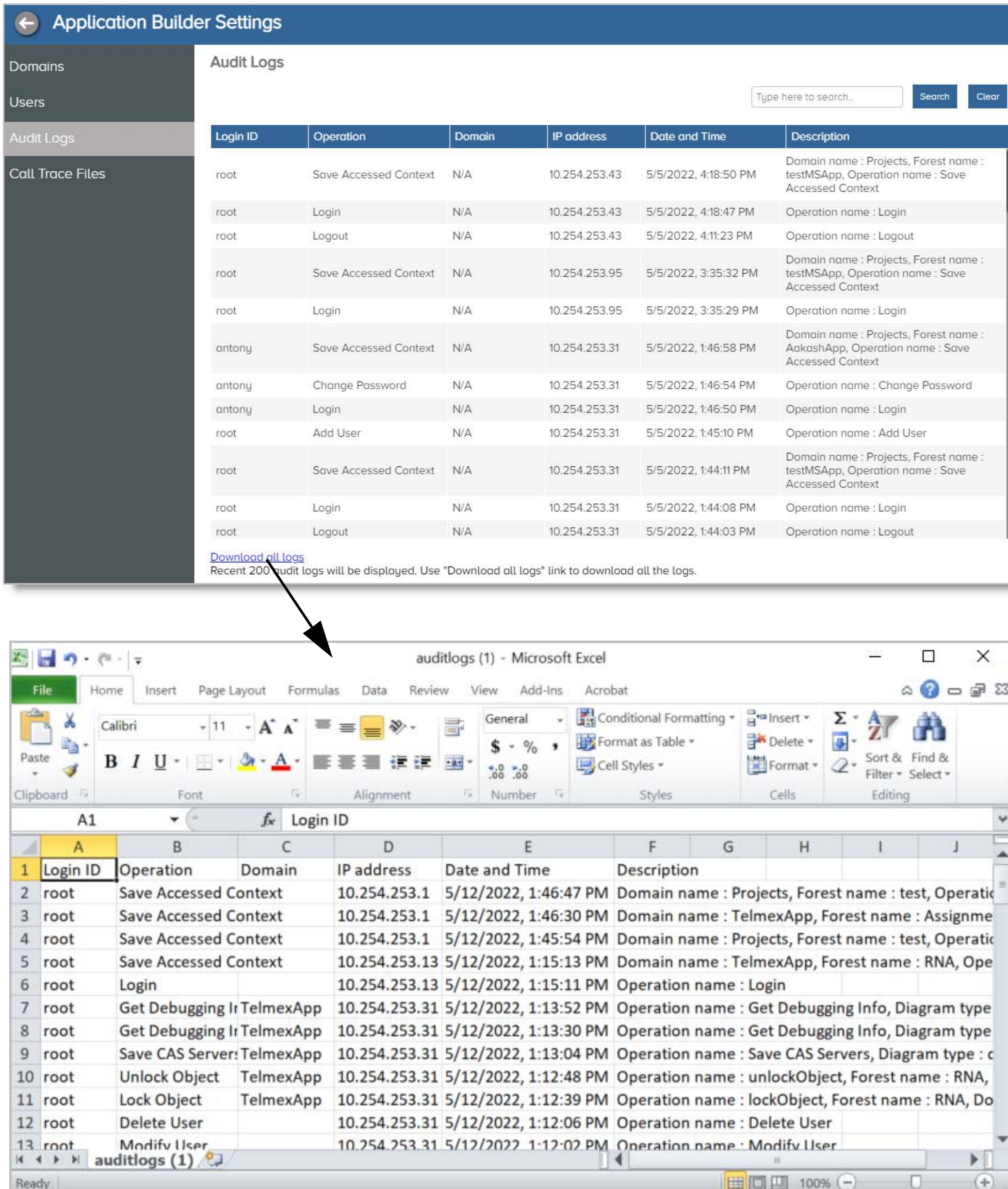


Figure 12 Downloading an Audit Report

To download an audit report, follow these steps.

1 Access the Audit Logs page. Refer to View Audit Report.

2 Click **Download all logs** present at the bottom of the page. Refer to Figure 12, Downloading an Audit Report.

The audit report gets downloaded in the csv format.

Tracing Calls

Call tracing is a diagnostic facility to visualize and debug call related problems in the Server deployment. It lets you trace calls by running test calls thereby capturing and analyzing their link data. Using the interface, you can view the history or progress of calls. You can debug the call related issues. For more details on debugging an application, refer Debugging an Application.

Figure 13 shows the pages used for Viewing Call Trace Details.

Application Name	Domain Name	Date and Time	Calling Party	Called Party	CAS Instance	Actions
231	System1	5/3/2022, 9:10:26 PM	1000000000	1000000000	cas_trainee	
231	System1	5/3/2022, 9:12:07 PM	1000000000	3300000000	cas_trainee	
231	System1	5/3/2022, 4:56:13 PM	1000000001	0000330000	cas_trainee	
231	System1	5/3/2022, 4:18:12 PM	1000000001	000100000000	cas_trainee	
231	System1	5/3/2022, 5:09:37 PM	1000000002	331000000078	cas_trainee	
231	System1	5/3/2022, 5:10:59 PM	1000000002	57	cas_trainee	
231	System1	5/3/2022, 5:25:24 PM	1000000002	57	cas_trainee	
231	System1	5/3/2022, 5:25:37 PM	1000000002	57	cas_trainee	
231	System1	5/3/2022, 5:25:39 PM	1000000002	57	cas_trainee	
231	System1	5/3/2022, 5:53:48 PM	1000000003	3310000000	cas_trainee	
231	System1	5/3/2022, 5:54:23 PM	1000000003	3310000000	cas_trainee	
231	System1	5/3/2022, 5:49:28 PM	1000000008	2210000000	cas_trainee	

Figure 13 Viewing Call Trace Details

Table 6 describes the Call Trace Parameters.

Table 6 Call Trace Parameters

Fields	Description
Application Name	Displays the application name.
Domain Name	Displays the name of the domain.
Date and Time	Displays the date and time of the traced call.
Calling Party	Displays the calling party number.
Called Party	Displays the called party number.
Instance	Displays the name of the instance that sends the call trace details.

To view details of the traced calls, follow these steps.

1 Log in to the interface. Refer to Accessing Application Development.

2 Click **Settings** icon present on the top-right corner of the screen. Refer to Figure 3, Application Development Interface. The Application Builder Settings page appears.

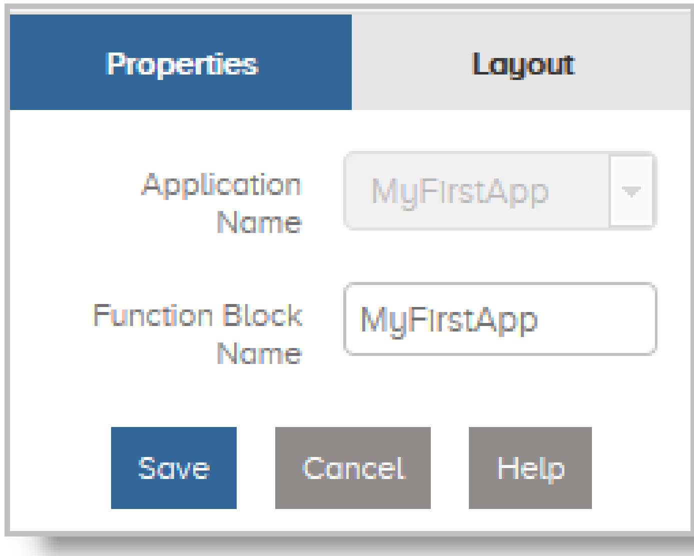
3 From the left menu pane, select **Call Trace Files**. The Call Trace Files page appears. Refer to Figure 13, Viewing Call Trace Details. For more details, refer to Table 6, Call Trace Parameters.

The call trace data that becomes visible on the screen.

When a call trace record is not required, then identify the particular record and click **Delete** icon. The identified record gets deleted from the list.

Understanding the Application Building Blocks

The provides a lot of flexibility and ease for creating applications. Before you start creating an application, you should understand the application building blocks.



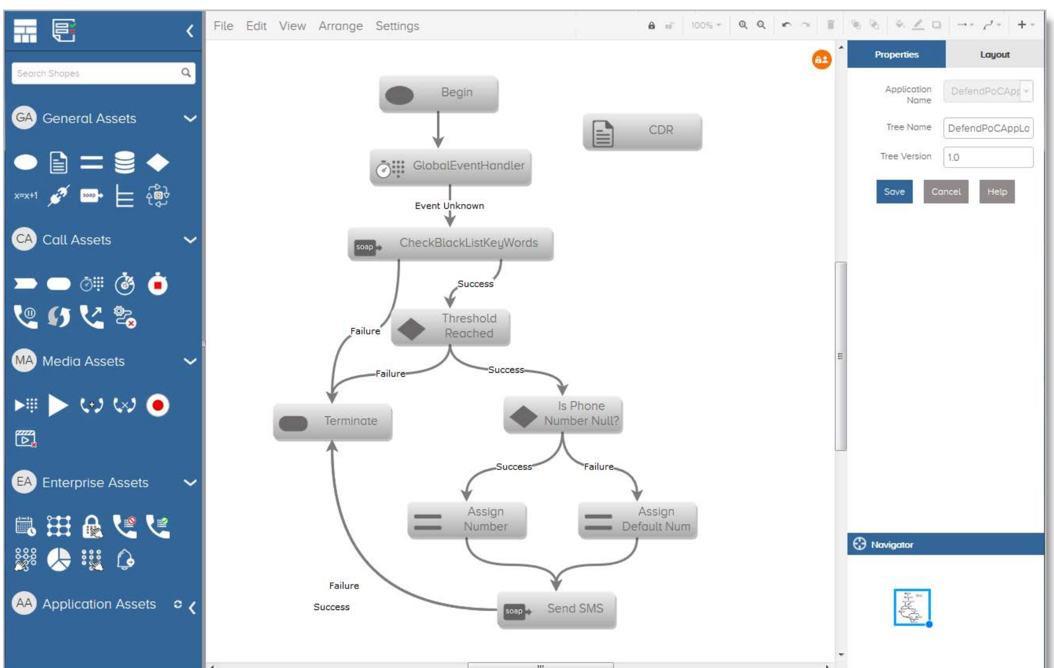


Figure 14 Application Building Blocks

An application can contain the following building blocks:

Functional Blocks: An application can contain a single or multiple functional blocks. Each functional block contains a logical grouping of business logic. For instance, an application may handle traffic from different network elements in a different way depending upon the business logic for that network. It can contain sub functions and trees.

Trees: Trees contain business logic created by connecting different assets. Your business logic can contain single or multiple trees. It is possible to jump from one tree to another tree. You can define your business logic into multiple trees depending on the functionality. Business logic may comprise of multiple trees where each tree may contain sub logic of an application.

Assets: Assets are the basic building blocks with specific functionality and associated properties used by an application. Assets can be connected logically to achieve desired business behavior of an application. Assets are connected using connectors (arrows) which decide the flow of call execution within the application. Depending upon the result of execution of that asset, you can branch out to next assets in case of success or failure. For some assets, errors are further classified for granularity and handling. For example in case of the DB Query node, the failure could be due to connection failure, result not found, or DB exception. You may either club the failure handling to common or may decide to perform different action (next asset) for different error cases.

Understanding Variables

Application Development supports various type of variables for holding values and are used for call processing. You can define your own variables or use the pre-defined variables in your application logic. These variables can contain both static and variable data.

Provides the following variables:

SMS variables: These variables are defined in *<sms-value>* which signifies that the variable value is set during subscriber provisioning.

Local variables: The value in these variables is set either during initialization or while running the logic such as database query response. You can define the local variables within the *setVar* tag.

Global variables: These are pre-defined variables and are stored in the call data. Global variables start with GBL. Global variables are populated by application from the incoming signaling message.

LEG variables: These variables are fixed and start either with LEG1_ or LEG2_.

CALLDATA variables: These variables are fixed and start with CALLDATA_. These variables contain value specific to each call.

CHAPTER 3

Developing Applications

This chapter provides information on how to create, debug, and deploy the applications created using Application Development.

Topics

This section includes the following topics:

Overview

The first step to develop an application using Application Development is to create an application by providing the application name. The application contains various building blocks such as functional blocks, trees, and functional nodes. For more details, see section Understanding the Application Building Blocks.

Before creating an application, you should be ready with the application logic and inputs required to create an application. For basic steps to create an application, see section Using Application Development.

Creating an Application

You can create, save, open, and delete the applications created using Application Development.

Figure 15 shows the procedure of creating a new application.

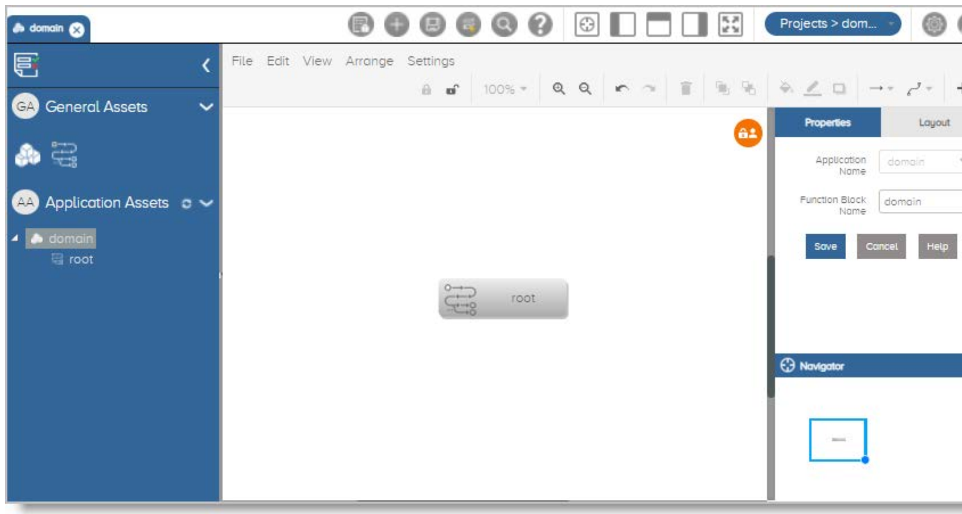


Figure 15 Creating a New Application

To create an application, follow these steps:

1. Access the interface. For details, see section Accessing Application Development.
2. In Application Development, go to the Title bar and click **Application** drop-down arrow button. For details, see Table 2, Title Bar Icons and Description.
3. Click **+New Application**. The Create Application window displays. See Figure 15, Creating a New Application.
4. Enter the application name. See Figure 15, Creating a New Application.

5 Click **Add**.

A new application is created.

Developing Applications

To develop an application, you must add the required application components from the application pallet and define application logic. The project and its components appear in the Application Pallet of Application Development.

This section includes the following application development tasks.

Add a Function Block

Add a Tree

Add an Application Asset

Connect Application Components

Adding Text Labels

Set Application Component Properties

Save an Application

Validate an Application

Add a Function Block

You can logically group different features of an application into logical units referred to as Function Blocks. A function block may or may not contain one or more function blocks. The Main Logic function block may or may not contain sub-groups. However, a function block at the lowest hierarchy must contain one or more tree nodes. You can also add a function block within a function block.

Figure 16 shows the proceduree of adding a new Function Block.

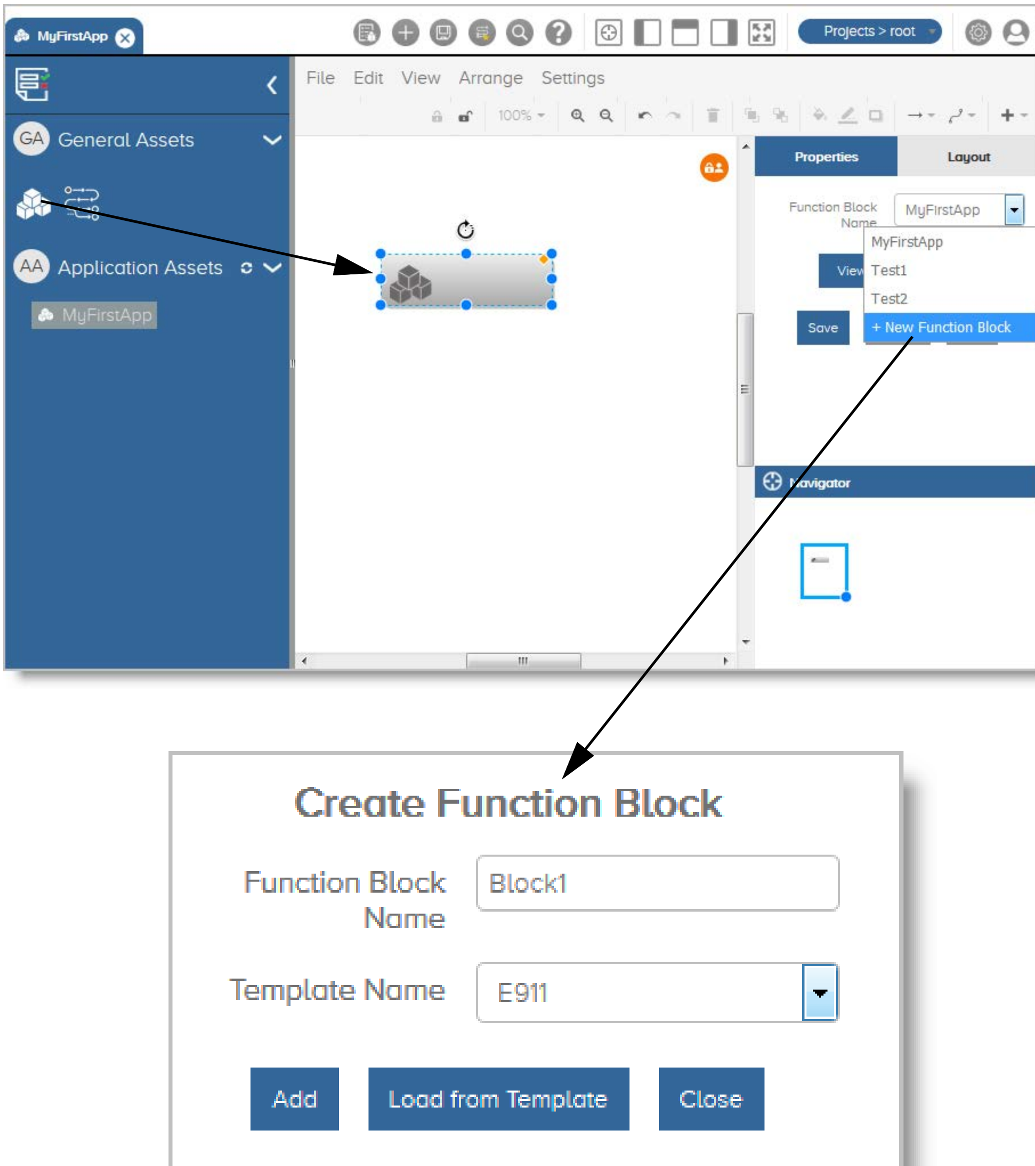


Figure 16 Adding a New Function Block

To add a new Function Block to your application, follow these steps:

1. Open the application.
2. From the Application Pallet (left pane), drag and drop the Function Block asset into the canvas. For details, see Table 1, GUI Components of the Application Development Interface.
3. In the Properties pane, click the **Function Block Name** drop-down arrow button. See Figure 16, Adding a New Function Block.
4. Click **+New Function Block**. The Create Function Block window displays. See Figure 16, Adding a New Function Block.
5. Enter the Function Block name. See Figure 16, Adding a New Function Block.
6. Click **Add**.

A new function block is added to your application.

Add a Tree

Figure 17 shows the procedure of adding a new Tree.

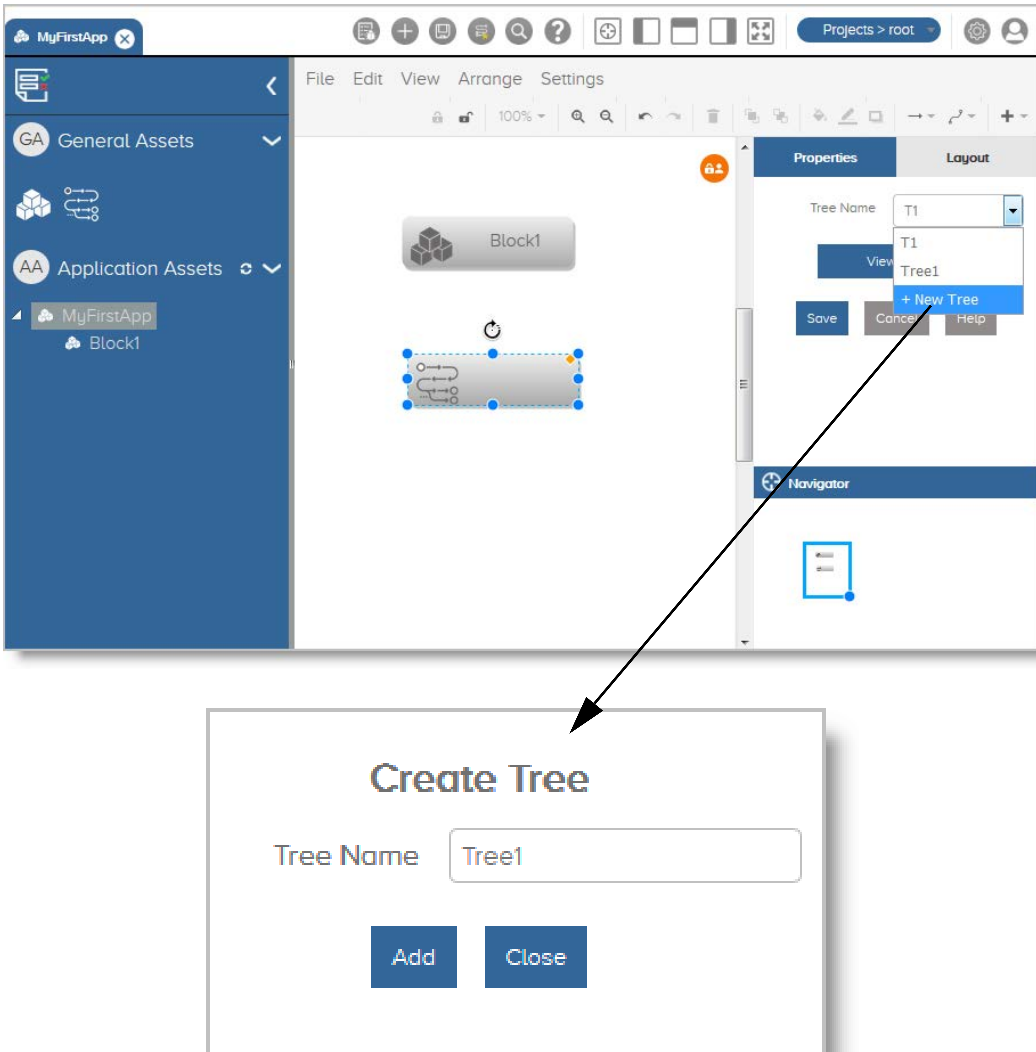


Figure 17 Adding a New Tree

To add a new Tree to your application, follow these steps:

1 Open the application.

2 From the Application Pallet (left pane), drag and drop the Tree asset into the canvas. For details, see Table 1, GUI Components of the Application Development Interface.

3 In the Properties pane, click the **Tree Name** drop-down arrow button. See Figure 17, Adding a New Tree.

4 Click **+New Tree**. The Create Tree window displays. See Figure 17, Adding a New Tree.

5 Enter the Tree name. See Figure 17, Adding a New Tree.

6 Click **Add**.

A new tree is added to your application.

Add an Application Asset

Assets are the actual building blocks of the application. You can add different assets to a tree to build the application logic and join them. The general procedure to add an asset to your application is the same as that of adding a functional block or tree.

Figure 18 shows the procedure of adding an asset in the application Tree.

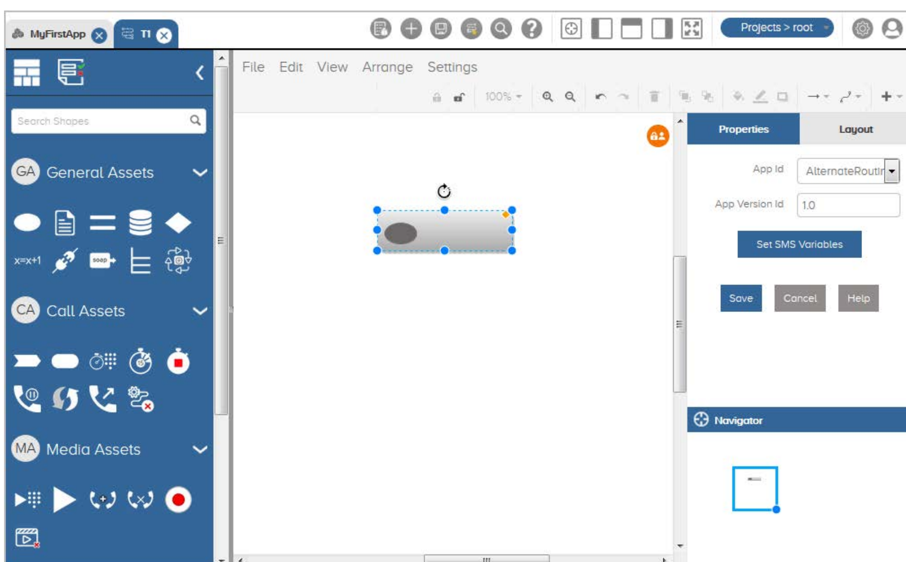


Figure 18 Adding an Asset

Table 7 describes application assets of Application Development.

Table 7 Application Assets

Component Name	Description
General Assets	
Start Node	Lets you begin the application logic. It is the starting point in an application logic execution. There must be only one Start node in each application.
Save CDR Node	Lets you define the fields of the CDR file.
Assignment Node	Lets you assign a value in a variable.
DB Call Node	Lets you run a stored procedure or a database query.
Condition Node	Provides conditional functions to be performed between two fields which could either variable or literal (static value).
Expression Node	Lets you create a mathematical expression between two or more fields which could either be variables or literals.
Connect Node	Lets you connect to another node either within the same tree or another tree of the same application.
External Call Node	Lets you invoke external JAVA program or SOAP API request.
Switch Node	Lets the user create switch case on a particular variable for conditional handling.
Process Call Node	Lets you add common functions (referred to as processes) which can be called from multiple trees.
Call Assets	
Route Call Node	Lets you route the SIP/SS7 calls.
Terminate Node	Lets the user disconnect call legs.
Event Handler Node	Listens to the mid-call events.
Create Timer Node	Lets the user create an application timer.
Stop Timer Node	Lets you stop the timer.
Call Hold Node	Lets you to put the ongoing call on hold. This node is applicable for SIP calls.

Table 7 Application Assets

Component Name	Description
Resync Call Node	Lets you to reconnect the caller and called party once put on hold. This node is applicable for SIP calls.
End Execution Node	Let you terminate the execution.
Automatic Code Gapping Node	Lets you check whether the number of calls exceeded the threshold value.
Media Assets	
Play and Collect Node	Lets the user listen to the announcements and then collect the choice entered by the caller/callee based on the connected party. The user needs to define the announcement depending upon the type of protocol like .wav file in case of SIP or announcement id in case of SS7.
Play Node	Lets you play the announcements either through media server in case of SIP or through switch in case of SS7.
Create Conference Node	Lets you add the participants to a conference call. This asset is valid for SIP calls only.
Destroy Conference Node	Lets you end a conference call. This asset is valid for SIP calls only.
Record Node	Lets the user to connect the caller to the media server to record information such as caller name. This node is applicable for SIP protocol.
Stop Media Node	Lets you stop an ongoing media interaction with the media server.
Send Email Node	Lets you send the email notifications.
Send SMS Node	Lets you send an SMS.
Enterprise Assets	
Send Alarm Node	Lets you specify the alarm code.
Routing Engine Node	Lets you execute the dynamically created routing logic through SPS.

Table 7 Application Assets

Component Name	Description
RE Route Play Node	This node is an extension of the Play node and is used in case a user selects the Advance RE Handler as Play. After playing an announcement, the call execution returns back to the Routing Engine node to execute the next Routing Engine node defined through SPS. No specific connectors are required to connect the RE Route Play node to the next node.
RE Route Play Collect Node	This node is an extension of the Play Collect node and is used when the user selects Advance RE Handler as Play Collect. After collecting the digits, the call execution returns back to the Routing Engine node to execute the next Routing Engine node defined through SPS. No specific connectors are required to connect the RE Route Play Collect node to the next node.
RE Route Node	Rename of the RE Route Node. The call execution returns back to the next node defined in the Routing Engine due to a routing failure such as no answer or busy error. No specific connectors are required to connect the Route node to the next node.
TUI Node	Lets you write the call data directly to the database.

To add an asset in your application tree, follow these steps:

1. Open the application. For details, see section Opening an Application.
2. Open the Tree. For details, see section View the Tree.
3. From the Application Pallet (left pane), drag and drop the asset into the canvas. For details, see Table 1, GUI Components of the Application Development Interface.
4. From the properties pane, set the asset properties. See Properties Pallet.
5. Click **Save**.

The asset is added to the tree.

Connect Application Components

You can connect various application components with each other to create an application.

To connect an application component

1 Select the application component from the application pallet.

2 Drag and drop the selected component on to the canvas.

3 Connect the component by dragging the arrow displayed on the component. Application Development displays the Connector Properties in the menu bar.

4 Set the connector properties.

5 Click **Save.**

For example,

To add a Play Prompt component

1 Select the **Play Prompt** component from the application pallet.

2 Drag and drop the **Play Prompt** component on to the canvas. The new **Play Prompt** appears on the canvas.

3 Connect it to the **Start** block. For this, perform the following steps:

   a Click the arrow present at the corner of the **Start** block.

   b Drag the arrow over the **Play Prompt** block and drop it.

The Play Prompt component is added.

Adding Text Labels

You can add text comments on the canvas for explaining the application logic.

To add text on the canvas, follow these steps:

1 Double-click your mouse pointer on the canvas where you need to add the text.

2 Type in your text in the text box.

The text becomes visible on the canvas.

Set Application Component Properties

Before you create an application you must set the properties of application components that are used for creating application call flow logic.

To set the properties of the application components, follow these steps:

1. Select the application component from the application pallet.
2. Drag and drop the selected application component on to the canvas. The properties window of the respective call component appears.
3. Set the properties as required. For details on properties of the application components, see Appendix A, Application Components.
4. Click **Save** to apply the settings.

The application component properties are set.

Save an Application

It is recommended that you keep saving your application from time to time. To save an application, go to the Title bar and click at the Save button.

The function block becomes visible in the Application Pallet only after you save it in the application. To save the function block, go to the Title bar and click at the Save button. See Figure 19, Saving an Application.

Figure 19 shows the procedure of saving an application.

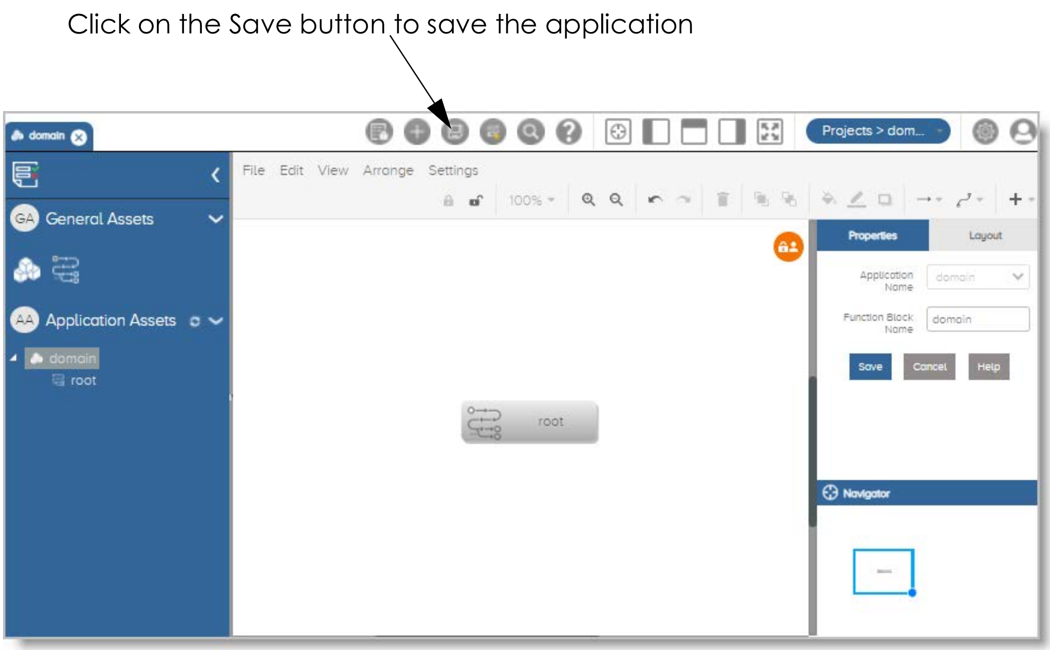


Figure 19 Saving an Application

Validate an Application

Once the application is deployed and the call is executed, it writes node execution string in the test logs. The string (identified with "XML Logic:") can be provided to trace the node execution through debug node.

To debug an application

1 Open the application that needs to be debugged. For instructions, see section Opening an Application.

2 From the Title bar, click the Debug icon. See Title Bar Icons and Description.

Lock an Application

Multiple users can simultaneously work on the applications. You can lock your application or tree while you are working on it to avoid any conflicts. This will enable only a single user to work on the application at any given point of time. After you have finished making the changes to your application or tree, ensure that you unlock it so that the other users can also work on it. For more details, see Unlock an Application.

Figure 20 shows the pages for Locking an Application.

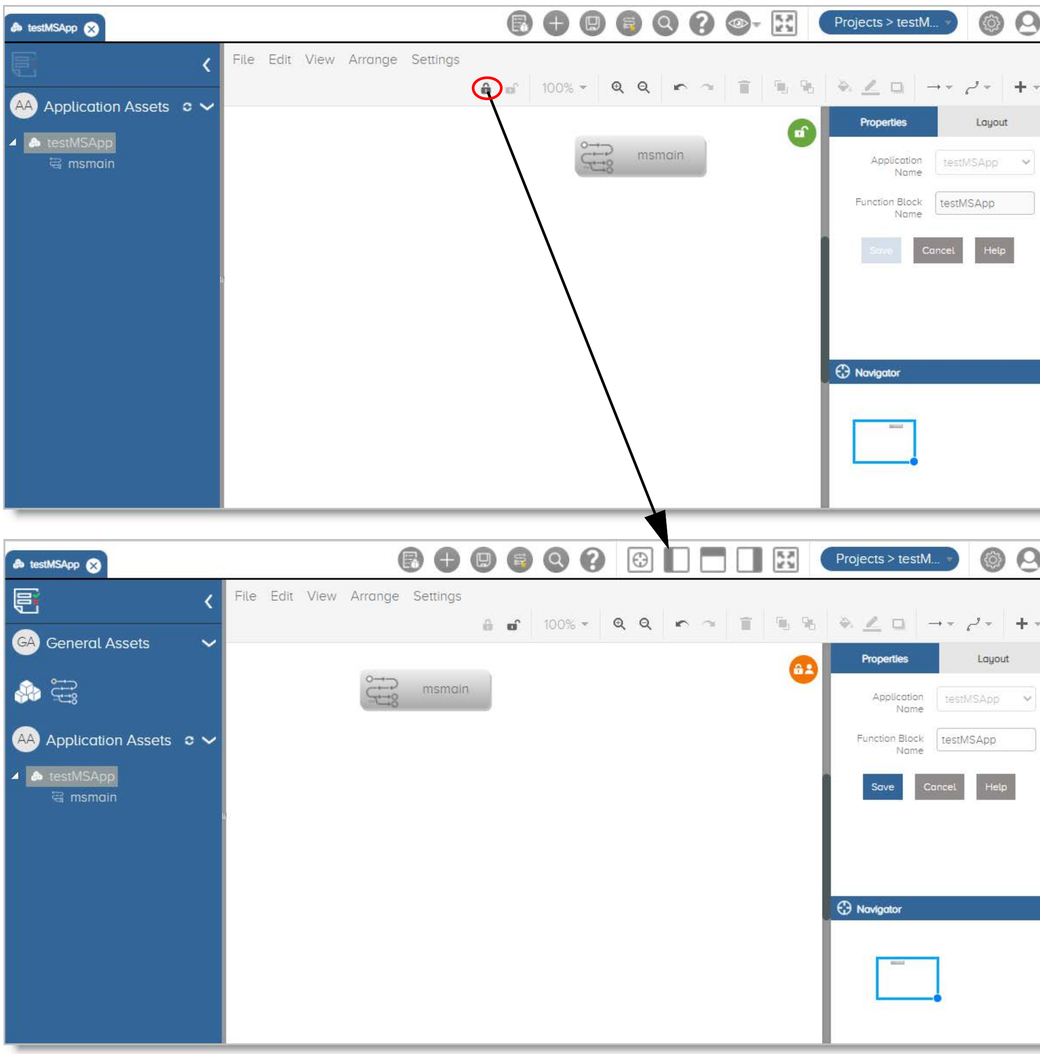


Figure 20 Locking an Application

To lock an application, follow these steps.

1. Log in to the interface.
2. Open the application. Refer to Opening an Application.
3. Click on the Lock icon. Refer to Figure 20, Locking an Application.

The application gets locked.

Unlock an Application

You should unlock the application or tree once you have finalized it. This will enable other users to make changes in the application whenever required.

Figure 21 shows the pages for Unlocking an Application.

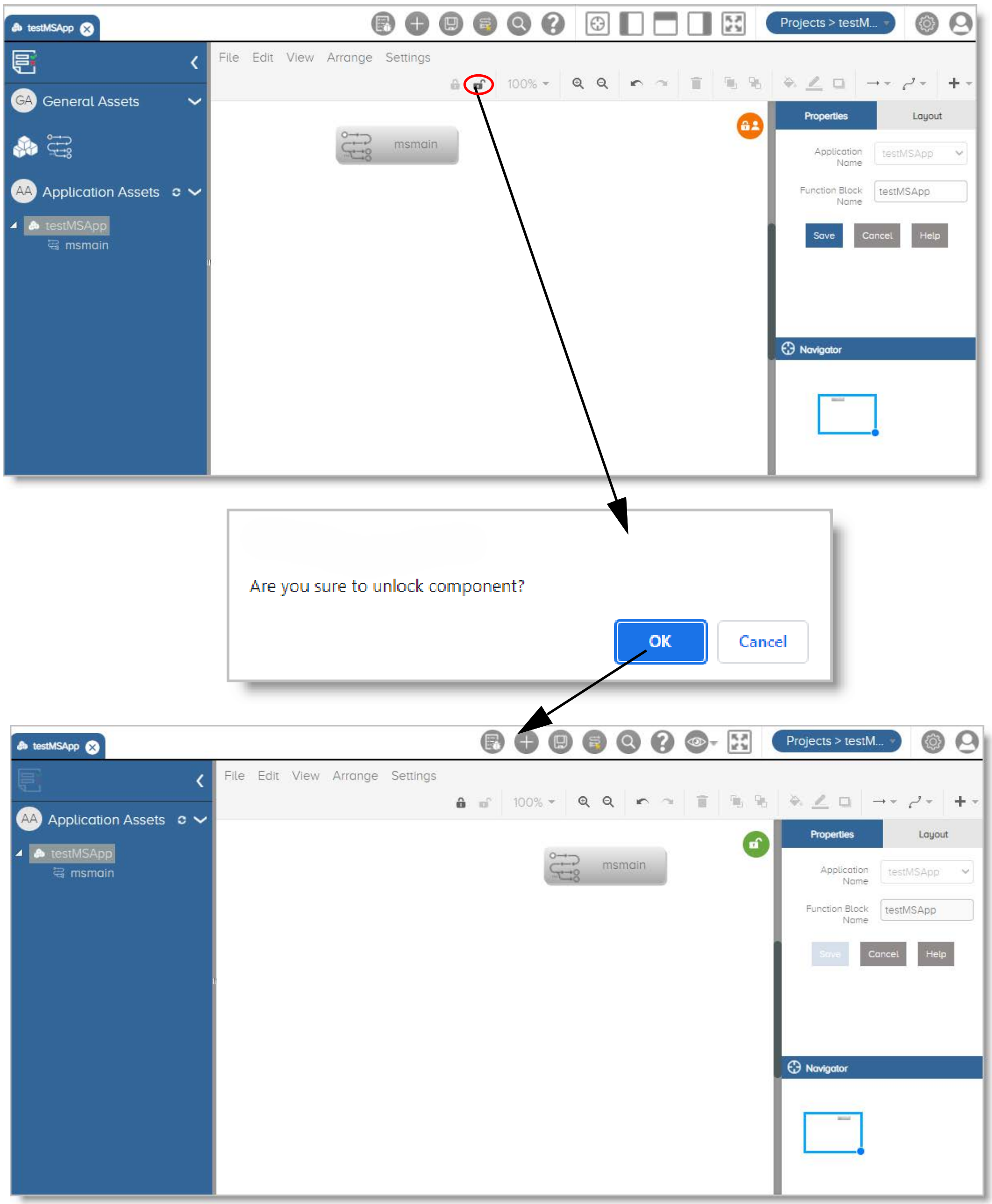

testMSApp
Projects > testM...
File
Edit
View
Arrange
Settings
100%
GA
General Assets
AA
Application Assets
testMSApp
msmain
msmain
Properties
Layout
Application Name
testMSApp
Function Block Name
testMSApp
Save
Cancel
Help
Navigator
Are you sure to unlock component?
OK
Cancel
testMSApp
Projects > testM...
File
Edit
View
Arrange
Settings
100%
AA
Application Assets
testMSApp
msmain
msmain
Properties
Layout
Application Name
testMSApp
Function Block Name
testMSApp
Save
Cancel
Help
Navigator

Figure 21 Unlocking an Application

To unlock an application, follow these steps.

1. Log in to the interface.
2. Open the application. Refer to Opening an Application.
3. Click on the Unlock icon. Refer to Figure 21, Unlocking an Application.

The application gets unlocked.

Closing an Application

To close an application, go to the Title bar and click at the cross button of the application name.

Figure 22 shows the procedure of closing an application.

Click on the cross button to close the application

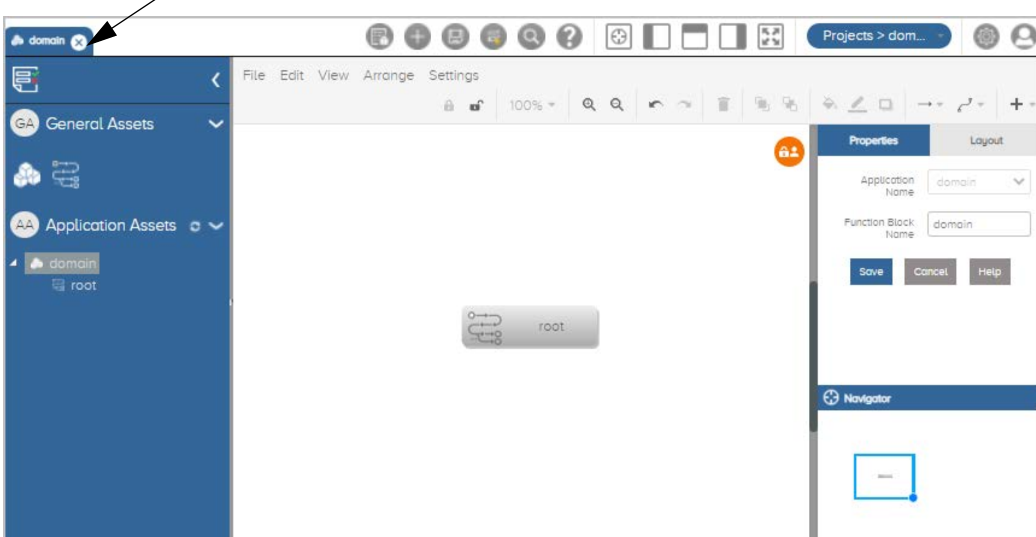


Figure 22 Closing an Application

Opening an Application

Figure 23 shows the procedure of opening an application.

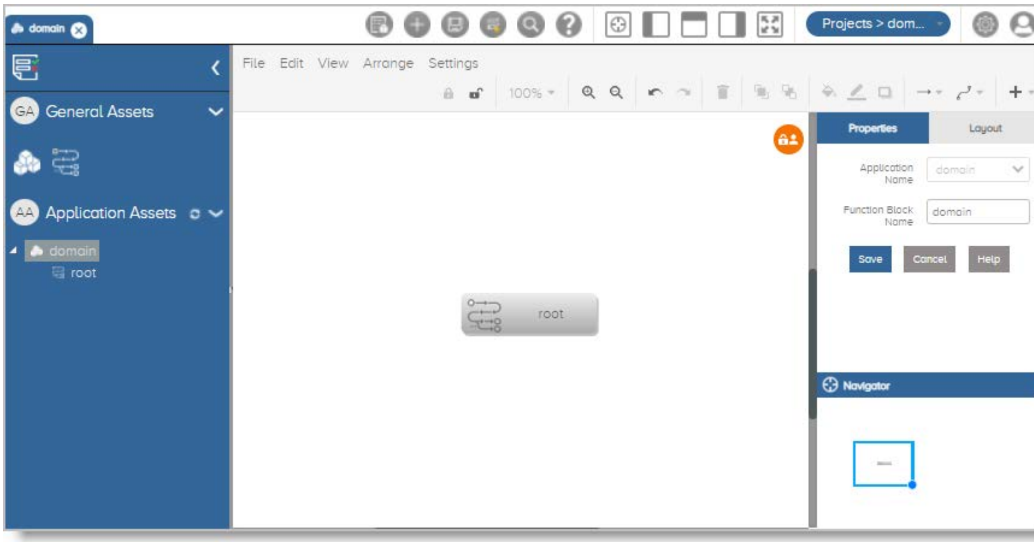


Figure 23 Opening an Application

To open an application, follow these steps:

1. Access the Application Development interface. For details, see section Accessing Application Development.
2. In Application Development, go to the Title bar and click **Application** drop-down arrow button. For details, see Table 2, Title Bar Icons and Description
3. Select the application name to open. The Select Type of Xml To Proceed window displays. See Figure 23, Opening an Application.
4. Select any one of the following radio buttons:

    User Saved Xml

    Auto Saved Xml
5. Click **Load**.

The application is opened.

Deleting an Application

To delete an application, follow these steps:

1. Open the application to delete. For details, see section Opening an Application.
2. In Application Development, go to the File menu and click **Delete**. For details, see Table 2, Title Bar Icons and Description.

   The Delete Application/Tree dialog box appears.

3. Select the application name.
4. Click **Delete Application**.

The application is deleted.

Managing an Application

You can create, save, open, and delete the applications created using Application Development.

This section includes the following topics:

View a Function Block

Add a Tree

View the Tree

Delete a component

View a Function Block

You can view the functional block and make changes to it.

To open a Function Block, follow these steps:

1. From the canvas, select the function block to open.
2. In the Properties pane, select **View Function Block**.
3. Select any one of the following radio buttons:

   User Saved Xml
   Auto Saved Xml

4. Click **Load**.

The Function Block opens in a new tab.

Add a Tree

You can add a tree directly in your application or within a functional block. The general procedure to add a tree to your application is the same as that of adding a functional block.

To add a tree

1. Drag and drop the tree component on to the canvas.
2. In the Properties window, click at the Tree Name drop down arrow.
3. Select **+New Tree**. The Create Tree dialog box appears.
4. Click **Add**.
5. Click **Save**.

The new tree is added to the application.

View the Tree

You can view the tree and make changes to it.

To view a tree, follow these steps.

1. Select the tree component from the canvas that you want to view.
2. In the Properties window, click at the **View Tree** button.

You can now view the tree contents and make changes to it.

Delete a component

You can delete an unwanted functional block, tree, or node at any time.

To delete a component, follow these steps:

1. Select the component name from the canvas.
2. Right click your mouse button and select **Delete**.

The component gets deleted from the canvas.

Deploying an Application

You can deploy your application on the server.

Table 8 describes the Application Details.

Table 8 Application Details

Parameter	Description
Available Servers	Select the name from the drop-down list where the application needs to be deployed.
Application Name	Displays the application name.
Priority	Defines the priority in which the application should be loaded. Multiple applications can share the same priority. You can define the priority from 1 to 5.

To deploy an application, follow these steps.

1 In Application Development, go to the menu bar and click **Settings**.

2 Select **Manage Server**. The Manage Server Applications page appears.

3 Enter the application deployment details. For details, see Table 8, Application Details.

4 Click **Deploy Application**. A message appears confirming successful deployment of the service.

5 Click **Close**.

The Application Development deploys the application.

You can undeploy the application. To undeploy the application, identify the relevant application record from the list and click the **Delete** icon.

Debugging an Application

Once your application is deployed, you can debug the application from the logs to trace the node execution through debug node. Before debugging the application, ensure that the application debug setting is enabled. For details on enabling the debug setting, see Enable Application Debugging.

Figure 24 shows the procedure of debugging an application.

Figure 24 Debugging an Application

To debug the application, follow these steps.

1 Open the application that needs to be debugged. For details, see section Opening an Application.

2 From the Title bar, click the Debug icon. See Figure 4, Title Bar Options.

3 Click **Poll New files** icon present on the Call Debug page. The call debug files get uploaded.

4 Click **Debug icon**.

5 Click on the **Play** button.

6 Click **Close** when finished.

The debug results are shown on the screen.

To upload call trace files, click the **Upload** Call Trace File icon present on the top right corner of the call debug page. Then choose the file and open it. The debug results appear.

Downloading Application XML

You can make modification to a single tree and can replace the XML. Alternatively, you can also repackage the application and deploy again.

Note: It is not recommended to perform this activity.

To download the specific tree, follow these steps.

1 Open the Tree whose application XML needs to be downloaded. For instructions, see section View the Tree.

2 From the Menu bar, click **File**.

3 Select **Download Service XML**. In the dialog box, specify the location of the file and click **Save**.

Working with Templates

You can save a specific tree logic as template. Template is a reusable logic which you can import in any other application. It provides re-usability across applications.

Import a Template

You need to first open the application, function block, or tree in which you need to add the logic from the template.

Figure 25 shows the procedure of importing a template.

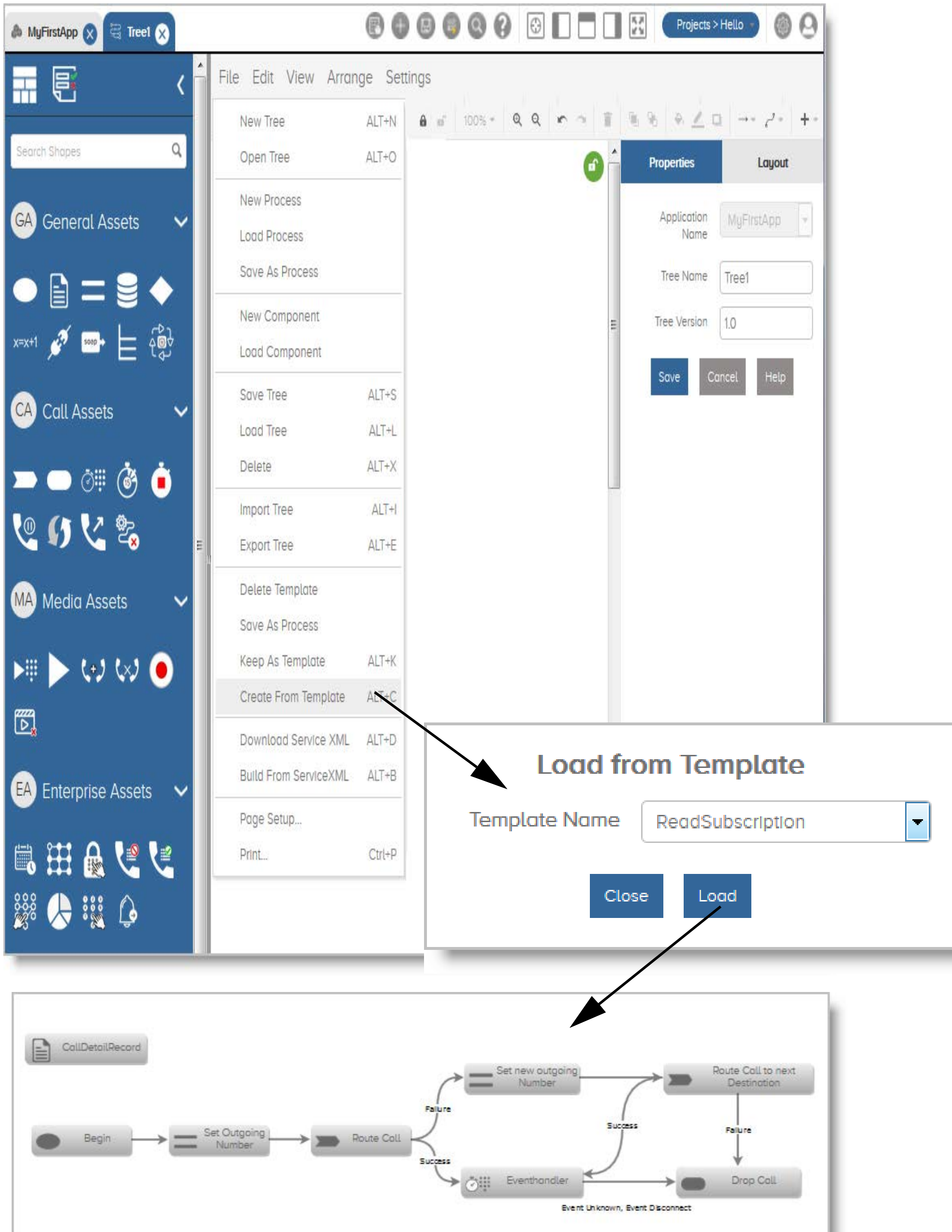


Figure 25 Importing a Template

To import a template, follow these steps.

1. Open the application, function block, or tree where you need to import the template contents.
2. In Application Development, go to the menu bar and click **File**.
3. Click **Create From Template**. The Load From Template dialog box appears.
4. Select the template name from the drop down menu.
5. Click **Load**. The template logic gets imported in your application.
6. Make the changes, if required.
7. Save your application.

Save a Template

You can save an application, functional block, or tree as a template and use it in future for building new applications.

To save your business logic as a template, follow these steps.

1. Create the application, function block, or tree.
2. In Application Development, go to the menu bar and click **File**.
3. Click **Keep As Template**. The Save as Template dialog box appears.
4. Type in the name of the template file.
5. Click **Save**.

The template file is saved.

Delete a Template

You can delete the templates related to function blocks and trees from the Application Development interface.

Delete a Function Block Template

To delete a function block template, follow these steps.

1. In Application Development, go to the menu bar and click **File**.
2. Click **Delete Template**. The **Delete Function Block Template** dialog box appears.
3. Select the name of the template file from the drop-down menu.
4. Click **Delete Template**.

The selected template file is deleted.

Delete a Tree Template

To delete a tree template, follow these steps:

1. Open any tree.
2. In Application Development, go to the menu bar and click **File**.
3. Click **Delete Template**. The **Delete Tree Template** dialog box appears.
4. Select the name of the template file from the drop-down menu.
5. Click **Delete Template**.

The selected template file is deleted.

Working with Processes

Processes refer to functions. Each process has its own input and output parameters. Process is a centralized logic that can be called from different places in a tree and has its own local variables.

Create a Process

A function block is just a logical grouping whereas a process is a function which implements common logic. You can run a process by calling it from different places in a tree.

Open a Process

Figure 26 shows the procedure of opening a process.

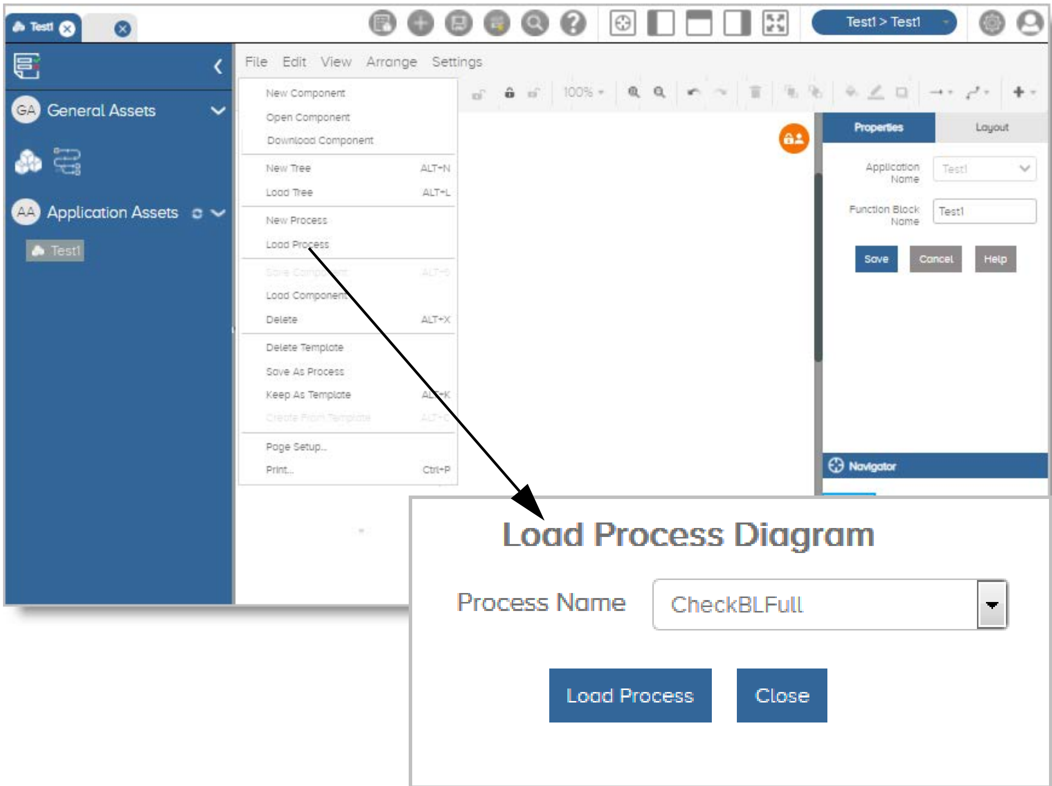


Figure 26 Opening a Process

To open a process, follow these steps.

1. In Application Development, go to the menu bar and click **File**.
2. Click **Load Process**. The **Load Process Diagram** dialog box appears.
3. Select the process name from the drop-down menu.
4. Click **Load Process**.
5. Select any one of the following radio buttons:

   User Saved Xml
   Auto Saved Xml

6 Click **Load**.

The process is opened and becomes visible in the canvas.

Export a Process

You can export your process in the XML format. The exported file is saved with the **.diagram** extension at the specified location.

Figure 27 shows the procedure of exporting a process.

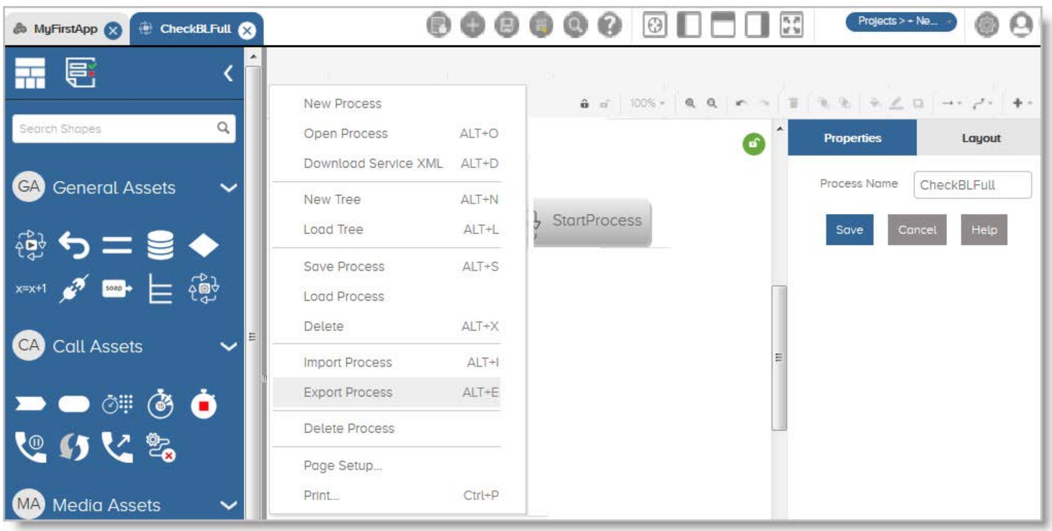


Figure 27 Exporting a Process

To export a process, follow these steps.

1 Create the process.

2 In Application Development, go to the menu bar and click **File**.

3 Click **Export Process**.

The process file gets downloaded. Then select an application to open the file.

Import a Process

To import a process, follow these steps.

1. Open the process in which you need to import the process file.
2. In Application Development, go to the menu bar and click **File**.
3. Click **Import Process**.
4. Click **Browse.**
5. Navigate to the location where the process file is saved.
6. Click **Open**.
7. Click **Import Diagram**.

The process file is imported in your application.

Delete a Process

To delete a process, follow these steps.

1. In Application Development, go to the menu bar and click **File**.
2. Click **Delete Process**. The **Delete Process** dialog box appears.
3. Select the process name from the drop-down menu.
4. Click **Delete Process**.

The selected process is deleted.